MS-12 (28-Dec-2025)

ANSWER KEY • Q 1–180 • Physics | Chemistry | Biology

PHYSICS (Q 1 – 45)

1	2	3	4	5	6	7*	8	9	10	11	12	13	14*	15
3	3	2	4	2	3	4	4	4	4	4	2	2	4	2
16	17	18	19	20	21	22*	23	24	25	26	27	28	29	30
4	2	1	3	4	2	4	4	1	2	4	3	1	3	1
31	32	33	34	35	36*	37	38	39	40	41	42	43	44	45
2	1	3	3	1	2	3	3	1	2	3	2	4	2	3

CHEMISTRY (Q 46 – 90)

46	47	48	49	50	51	52	53*	54	55	56	57	58	59	60
4	3	4	1	1	4	3	3	1	3	4	1	1	2	1
61	62	63	64	65	66	67	68	69	70*	71	72	73	74	75
3	3	2	3	3	3	2	4	3	2	4	1	1	2	1
76	77	78	79	80	81	82*	83	84	85	86	87	88	89	90
2	2	4	4	1	3	2	4	3	4	2	3	4	4	1

BIOLOGY (Q 91 – 180)

91	92	93	94	95	96*	97	98	99	100	101	102	103	104	105
2	1	4	1	4	3	1	4	3	3	3	3	1	4	1
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
2	1	1	4	4	1	4	4	3	1	2	3	3	3	2
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
2	2	1	1	1	4	2	1	1	4	2	2	3	1	1
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
3	4	2	3	2	2	4	2	2	3	4	4	2	3	3
151	152	153	154	155	156	157	158*	159	160	161	162*	163	164	165
3	4	4	3	3	2	4	3	2	4	3	3	2	2	4
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
4	3	3	2	4	4	1	3	3	3	2	1	4	4	3

Note: Figure wale kuch questions mein diagram text mein clear nahi tha. Unke aage * lagaya hai (solutions mein [?]): **7, 14, 22, 36, 53, 70, 82, 96, 158, 162**. Inhe apni official key se zaroor match kar lena.

PHYSICS (Q 1–45)

Q1. (3) $q = \int_0^{10}(2+3t)\,dt = 2(10) + \frac{3(10)^2}{2} = 20 + 150 = \mathbf{170\,C}$

Q2. (3) $I = \frac{ne}{t} = 10^{20} \times 1.6 \times 10^{-19} = \mathbf{16\,A}$

Q3. (2) $v_d = \frac{I}{neA}$, $A \propto r^2$. $I \times 2$ aur $r \times 2 \Rightarrow A \times 4$. $v' = \frac{2}{4}v = \mathbf{v/2}$

Q4. (4) $P = \frac{V^2}{R}$, R constant $\Rightarrow \frac{P}{P_0} = \left(\frac{V}{V_0}\right)^2 \Rightarrow \mathbf{P = \left(\frac{V}{V_0}\right)^2 P_0}$

Q5. (2) V–I graph ka slope $= R$. $\frac{R_1}{R_2} = \frac{\tan 45^\circ}{\tan 30^\circ} = \frac{1}{1/\sqrt{3}} = \mathbf{\sqrt{3} : 1}$

Q6. (3) Volume constant, $R \propto L^2$. $L \times 2 \Rightarrow R' = \mathbf{4R}$

Q7. (4) [?] B_3 seedha 250 V par hai, to $W_3 = 60$ W (full). B_1 aur B_2 series mein hain. Series mein $P = I^2R$, aur $R = V^2/P$, to $R_1 < R_2 \Rightarrow W_1 < W_2$. Aur $W_2 < 60$ W. Isliye $\mathbf{W_1 < W_2 < W_3}$

Q8. (4) KCL (poore network par): andar jane wali $12 + 3 + 5 = 20$ A $\Rightarrow$ $\mathbf{I = 20\,A}$

Q9. (4) Ring zero resistance hai. Center se ring tak teen $6R$ parallel $= 2R$. A tak ek aur $2R$ series mein hai. Total $= 2R + 2R = \mathbf{4R}$

Q10. (4) $\alpha = \frac{R_{100} - R_0}{R_0 \times 100} = \frac{7}{3 \times 100} = \mathbf{2.3 \times 10^{-2}\,^\circ C^{-1}}$

Q11. (4) $I_{\text{avg}} = \frac{\text{Area}}{\text{time}} = \frac{\frac{1}{2} \times 15 \times 10}{15} = \mathbf{5\,A}$

Q12. (2) Series: $R_{eq} = 2R$. Parallel: $R_{eq} < R$. Same V ke liye $H = \frac{V^2}{R_{eq}}$ per sec, jahan R_{eq} kam wahan zyada heat $\Rightarrow$ **Case-2 mein zyada**

Q13. (2) $i = \frac{6}{1+2+3} = 1$ A $\Rightarrow V_{3\Omega} = 1 \times 3 = \mathbf{3\,V}$

Q14. (4) [?] Teeno branches L aur R nodes ke beech parallel hain. EF branch (1 V, $r = 0$) se $V_L - V_R = 1$ V fix hota hai. CD branch mein $I = \frac{2-1}{3} = \mathbf{\frac{1}{3}\,A}$, **C se D** (cell polarity figure ke hisaab se same maani hai).

Q15. (2) $R = \frac{\rho L}{A}$. $R_{\max}$: $L = a_{\text{long}}$, $A = a_{\text{short}} \times a_{\text{mid}}$. $R_{\min}$: $L = a_{\text{short}}$, $A = a_{\text{long}} \times a_{\text{mid}}$. Ratio $= \left(\frac{a_{\text{long}}}{a_{\text{short}}}\right)^2 = \mathbf{4}$ (beech wala edge cancel ho jata hai)

Q16. (4) Har bulb $R = \frac{220^2}{100} = 484\ \Omega$. Series $R = 968\ \Omega$. $P = \frac{110^2}{968} = \mathbf{12.5\,W}$

Q17. (2) $P' = P\left(\frac{V'}{V}\right)^2 = 50 \times \left(\frac{1}{2}\right)^2 = \mathbf{12.5\,W}$

Q18. (1) Balanced bridge: $\frac{10}{15} = \frac{AD}{DB} \Rightarrow AD = 100 \times \frac{10}{25} = \mathbf{40\,cm}$ from A

Q19. (3) $R = 1 + \frac{2R}{2+R} \Rightarrow R^2 - R - 2 = 0 \Rightarrow \mathbf{R = 2\,\Omega}$

Q20. (4) Balanced Wheatstone bridge: $\frac{2}{x} = \frac{4}{8} \Rightarrow \mathbf{x = 4\,\Omega}$

Q21. (2) Discharge mein battery ke andar $+$ve charge $-$ se $+$ terminal ki taraf jata hai. Charging mein $+$ se $-$ ki taraf. Isliye "always" galat hai, sahi: **may go from $+$ to $-$**

Q22. (4) [?] Do cells opposite polarity mein parallel hain: $E_{eq} = \frac{\frac{4}{0.5} - \frac{2}{0.5}}{2+2} = 1$ V, $r_{eq} = 0.25\ \Omega$. $i = \frac{1}{3.75 + 0.25} = \mathbf{0.25\,A}$

Q23. (4) $I_{3\Omega} = 6 \times \frac{6}{6+3} = \mathbf{4\,A}$

Q24. (1) $R = \dfrac{V^2}{P} = \dfrac{220^2}{100} =$ **484 Ω**

Q25. (2) Free electron ki random thermal motion hoti hai, to average speed $\neq 0$ (Statement I galat). Average velocity $= 0$ (Statement II sahi).

Q26. (4) $P \propto I^2$. $(0.75)^2 = 0.5625 \Rightarrow$ power 43.75% **kam** hota hai.

Q27. (3) Case I: $V_A - V_B = E + Ir$, Case II: $E - Ir$. Isliye Case I mein PD zyada. A, R dono true aur R, A ki sahi vyakhya hai.

Q28. (1) $6 \parallel 3 = 2\ \Omega$. Total current I. $3\ \Omega$ mein current $= \dfrac{2I}{3}$. $P_3 = \dfrac{4I^2}{9}(3) = \dfrac{4I^2}{3}$, $P_4 = 4I^2$. Ratio = **1 : 3**

Q29. (3) D par: andar $5 + 3 = 8$ A, E ki taraf 8 A jata hai. E par: $8 = 2 + i \Rightarrow$ **$i = 6$ A**

Q30. (1) $60°$ arc $= \dfrac{36}{6} = 6\ \Omega$, bacha hua $= 30\ \Omega$. Parallel $= \dfrac{6 \times 30}{36} =$ **5 Ω**

Q31. (2) Nodes a, b, c, d lekar solve karne par $V_b = V_c = \dfrac{10}{3}$ V (bridge balanced, beech wala 10 Ω current nahi leta). $I = \dfrac{5/3}{5} + \dfrac{5/3}{10} = \dfrac{1}{3} + \dfrac{1}{6} =$ **0.5 A**

Q32. (1) $i = \dfrac{5+15}{2+4+14} = 1$ A. A–IV. $V_A = 5 - 1 \times 2 = 3$ V $\Rightarrow$ B–II. $V_{14\Omega} = 14$ V $\Rightarrow$ C–I. $V_B = 15 - 4 = 11$ V $\Rightarrow$ D–III. **A-IV, B-II, C-I, D-III**

Q33. (3) $3 \parallel 6 = 2\ \Omega$, usse 2 Ω series: $2 + 2 =$ **4 Ω**

Q34. (3) Battery P aur Q ke beech hai. A–P: 5 Ω, A–Q: 10 Ω $\Rightarrow I = \dfrac{4}{15}$, $V_A - V_Q = \dfrac{8}{3}$. B–P: 10 Ω, B–Q: 5 Ω $\Rightarrow V_B - V_Q = \dfrac{4}{3}$. $V_A - V_B =$ **$\frac{4}{3}$ V**

Q35. (1) $3 \parallel 6 = 2\ \Omega$. $V = 6$ V $\Rightarrow i = 3$ A. $r = \dfrac{8-6}{3} =$ **$\frac{2}{3}$ Ω**

Q36. (2) [?] A aur D ek hi node hain, C aur B ek hi node hain. Teeno resistors X–Y ke beech parallel hain. Current high potential wale (D) se low (C) ki taraf jata hai $\Rightarrow$ **D se C** (battery polarity figure ke hisaab se).

Q37. (3) Har diamond mein $(1 + 1) = 2\ \Omega \parallel (2 + 2) = 4\ \Omega = \dfrac{4}{3}\ \Omega$. Chaar series mein: **$\frac{16}{3}$ Ω**

Q38. (3) AB (30) $\parallel$ (BC+CA $= 60$) $= 20\ \Omega$. $I = \dfrac{2}{20} =$ **$\frac{1}{10}$ A**

Q39. (1) Dono path 20 Ω, har path mein 1 A. Top mid ka potential $V_0 - 4$, bottom mid ka $V_0 - 16$. Reading = **12 V**

Q40. (2) $\rho = \dfrac{E}{J} = \dfrac{E \cdot A}{I} = \dfrac{100 \times 10^{-7}}{5} = 2 \times 10^{-6}\ \Omega \cdot \text{m} =$ **$2 \times 10^{-3}\ \Omega \cdot \text{mm}$**

Q41. (3) $R_A = \dfrac{110^2}{25} = 484\ \Omega$, $R_B = 121\ \Omega$. $I = \dfrac{220}{605} = 0.364$ A. A ka rated current $\dfrac{25}{110} = 0.227$ A, jo kam hai, to **sirf A fuse hoga**.

Q42. (2) KVL energy conservation par based hai, yaani electric force ki **conservative nature** par.

Q43. (4) $\dfrac{R_1}{X} = \dfrac{40}{60} \Rightarrow X = \dfrac{10 \times 60}{40} =$ **15 Ω**

Q44. (2) $3 \parallel 6 = 2\ \Omega$, total 4 Ω, $I = 3$ A. $V_{\text{parallel}} = 6$ V. $P_3 = \dfrac{36}{3} =$ **12 W**

Q45. (3) Series mein $P \propto R$ (zyada $R = 50$ W wala bulb zyada chamakta hai), parallel mein $P \propto \dfrac{1}{R}$. A, R dono true aur R, A ki sahi vyakhya hai.

CHEMISTRY (Q 46–90)

Q46. (4) $t_{1/2}$ independent of $[R]_0 \Rightarrow$ first order $\Rightarrow$ unit **s^{-1}**

Q47. (3) 99.9% $\approx 10\, t_{1/2} = 450$ min = **7.5 hours**

Q48. **(4)** $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \cdot \frac{T_2 - T_1}{T_1 T_2}$, $T_1 = 320$, $T_2 = 330$. $0.301 = \frac{E_a \times 10}{2.303R \times 320 \times 330} \Rightarrow$ $\boxed{E_a = 6.9 \times 32 \times 33 \times R}$

Q49. **(1)** $t_{99.9\%} = \frac{3 \times 2.303}{k} = 10 \times t_{1/2} \Rightarrow$ **10 times**

Q50. **(1)** $\log C_t = \log C_0 - \frac{kt}{2.303} \Rightarrow$ **time vs** $\log C_t$ straight line deta hai.

Q51. **(4)** A true (fractional order hota hai), R true, par R, A ki explanation nahi hai.

Q52. **(3)** Intersection par $[A] = [B]$. $A_0 - x = nx \Rightarrow x = \frac{A_0}{n+1}$, $[B] = nx = \boxed{\frac{nA_0}{n+1}}$

Q53. **(3)** **[?]** Complex reaction mein overall molecularity ka koi meaning nahi hota (I true). Rate determining step ki molecularity hi order batati hai (II ko true maana gaya).

Q54. **(1)** $\mu^3 = 27$ (30 K = 3 intervals of 10 K) $\Rightarrow \boxed{\mu = 3}$

Q55. **(3)** B sahi (A ki unit k jaisi hoti hai). C sahi (kam E_a = tez reaction). A galat (Arrhenius sirf elementary ke liye nahi hai). D aur E galat. $\Rightarrow$ **B and C only**

Q56. **(4)** Pure liquid ka "concentration" constant hota hai, isliye pressure se rate nahi badalta.

Q57. **(1)** Bahut kam T par $k \approx 0$, to $A \neq k$. $T = \infty$ ya fraction = 1 par $k = A$ hota hai. Isliye A ko rate constant **nahi** kaha ja sakta **very low** T par.

Q58. **(1)** Pseudo first order reaction bimolecular hoti hai, dono mein molecularity 1 nahi hoti. **Yeh similar nahi hai.**

Q59. **(2)** C aur A ban rahe hain, B aur D kam ho rahe hain. Stoichiometry 2 : 4 : 1 : 3 $\Rightarrow$ **B + 3 D $\longrightarrow$ 4 A + 2 C**

Q60. **(1)** $k = \frac{2.303}{60} \log \frac{100}{40} =$ $0.01528\ \text{min}^{-1}$. $t_{50} = \frac{0.693}{k} \approx$ **45 min**

Q61. **(3)** Activation energy = potential barrier paar karne ke liye zaroori minimum extra energy.

Q62. **(3)** Exp 1 aur 3 mein $[A]$ double karne par rate same $\Rightarrow$ A ka order 0. Exp 1 aur 4 mein $[B]$ double karne par rate $\times 8 \Rightarrow$ B ka order 3. $\boxed{\text{Rate} = k[B]^3}$

Q63. **(2)** II sahi ($E_a = 0 \Rightarrow k = A$). V sahi ($\ln k$ vs $1/T$ straight line). I, III, IV galat. $\Rightarrow$ **II and V**

Q64. **(3)** Rate $= -\frac{1}{2}\frac{d[B]}{dt} = -\frac{d[A]}{dt} = \frac{1}{2}\frac{d[C]}{dt} \Rightarrow$ C ke banne ki rate A ke kam hone ki rate se do guna hai.

Q65. **(3)** First order ke liye k ki unit $\boxed{\text{s}^{-1}}$

Q66. **(3)** $\log k = \log A - \frac{E_a}{2.303RT} \Rightarrow$ slope $= \boxed{-\frac{E_a}{2.303R}}$

Q67. **(2)** Catalyst E_a ko kam karta hai, isliye A false. R true.

Q68. **(4)** $10^{16} e^{-2000/T} = 10^{15} e^{-1000/T} \Rightarrow$ $e^{1000/T} = 10 \Rightarrow \boxed{T = \frac{1000}{2.303}\ \text{K}}$

Q69. **(3)** Zero order mein rate $= k \Rightarrow k$ ki unit $\text{mol L}^{-1}\text{s}^{-1}$. R, A ki sahi explanation hai.

Q70. **(2)** **[?]** B ke reference se rate (B ke banne ki rate) $= \frac{d[B]}{dt} = \boxed{-\frac{2}{3}\frac{d[A]}{dt}}$

Q71. **(4)** Zero order ka k sirf **temperature** par depend karta hai.

Q72. **(1)** Graph mein 2 peaks hain, to 2-step reaction. Pehla barrier zyada ooncha hai $\Rightarrow$ **step 1 slow**. B intermediate hai.

Q73. **(1)** $k_1 = \frac{0.693}{40} = 0.01733\ \text{s}^{-1}$. $k_0 = \frac{[R]_0}{2t_{1/2}} = \frac{1.386}{40} =$

$0.03465\ \text{mol dm}^{-3}\text{s}^{-1}$. Ratio = **$0.5\ \text{mol}^{-1}\ \text{dm}^3$**

Q74. **(2)** $\ln\frac{k_2}{k_1} = \frac{58300}{8.314}\left(\frac{1}{300}-\frac{1}{305}\right) = 0.383 \Rightarrow$ ratio = **≈ 1.46**

Q75. **(1)** $[R] = [R]_0 - kt$, $[R] = 0$ par **$t = \frac{[R]_0}{k}$**

Q76. **(2)** I: A ka order 1. II: total order 3 ($2^3 = 8$) $\Rightarrow$ B ka order 2. **Rate $= k[A][B]^2$**

Q77. **(2)** Volume aadha karne par concentration double. Forward $\propto [SO_2]^2[O_2] \Rightarrow \times 8$. Backward $\propto [SO_3]^2 \Rightarrow \times 4$. Isliye **forward $= 2\times$ backward**

Q78. **(4)** Collision frequency 2–3% badhti hai (A true). 10°C par rate double hoti hai (R true), par R, A ki explanation nahi hai.

Q79. **(4)** $t_{99} = \frac{2\times 2.303}{k}$, $t_{90} = \frac{2.303}{k} \Rightarrow$ **$x = 2$**

Q80. **(1)** $t = \frac{2.303}{k}\log 10 = \frac{2.303}{2.303\times 10^{-2}} =$ **$100\ \text{s}$**

Q81. **(3)** Cane sugar inversion mein water excess hota hai $\Rightarrow$ **pseudo first order**

Q82. **(2) [?]** Statement (2) sahi hai (bimolecular elementary $\Rightarrow$ order 2), aur (3) bhi sahi hai (zero order complex hoti hai). (1) aur (4) galat hain. Matlab 2 statements sahi hain $\Rightarrow$ option (2). Paper mein options aur question ka format thoda mismatch hai.

Q83. **(4)** Rate $= -\frac{1}{3}\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

Q84. **(3)** $t_{3/4} = \frac{2.303}{k}\log\frac{1}{1-3/4} =$ **$\frac{2.303}{k}\log 4$**

Q85. **(4)** Zero order: $\text{mol L}^{-1}\text{s}^{-1}$ (A–IV). Second order: $\text{L mol}^{-1}\text{s}^{-1}$ (B–III). Order 3/2: $\text{L}^{1/2}\text{mol}^{-1/2}\text{s}^{-1}$ (C–II). First order: s^{-1} (D–I).

Q86. **(2)** $-\frac{1}{2}\frac{d[HI]}{dt} = \frac{d[H_2]}{dt} = \frac{d[I_2]}{dt}$

Q87. **(3)** Rate $= k[N_2O_5] \Rightarrow [N_2O_5] = \frac{2.4\times 10^{-5}}{3\times 10^{-5}} =$ **$0.8\ \text{M}$**

Q88. **(4)** Order fractional bhi ho sakta hai, isliye "cannot be fractional" galat hai.

Q89. **(4)** $-\frac{1}{2}\frac{d[X]}{dt} = \frac{d[X_2Y]}{dt} \Rightarrow X$ ke kam hone ki rate X_2Y ke banne ki rate se **do guna** hoti hai.

Q90. **(1)** Endothermic mein products ki energy zyada hoti hai (I true). Molecularity slowest step se decide hoti hai, "fastest" nahi (II false).

BIOLOGY (Q 91–180)

Q91. **(2)** Primary follicle = primary oocyte + ek layer of granulosa cells.

Q92. **(1)** Growing parts (young stem, petiole) ko mechanical support **collenchyma** deta hai.

Q93. **(4)** Epiblema = roots ki **outermost epidermal layer** (root hairs yahin hote hain).

Q94. **(1)** Monocot root: 6 se zyada xylem (polyarch) $\Rightarrow$ A–II. Dicot root: 2–4 xylem $\Rightarrow$ B–III. Monocot stem: scattered bundles $\Rightarrow$ C–IV. Dicot stem: epidermal hair $\Rightarrow$ D–I.

Q95. **(4)** Cycle ki sabse chhoti phase **ovulatory phase** (~1–2 din).

Q96. **(3) [?]** Stomatal apparatus = guard cells + subsidiary cells + stomatal pore $\Rightarrow$ **B, C, D** (figure ke hisaab se).

Q97. **(1)** Dicot stem mein hypodermis (collenchymatous) ke neeche cortex ki **parenchymatous** cells, intercellular spaces ke saath.

Q98. **(4)** A, B, C sahi. D galat (mesophyll parenchyma se bana hota hai, sclerenchyma se nahi).

Q99. **(3)** Reticulate venation mein veins ki thickness alag hoti hai, isliye bundles

ka size bhi alag hota hai.

Q100. **(3)** External genitalia: mons pubis, labia majora/minora, hymen, clitoris ⇒ **A, C, D, E**. Vagina internal hai.

Q101. **(3)** Dicot root cortex = **thin-walled parenchyma**, intercellular spaces ke saath.

Q102. **(3)** Water stress mein bulliform cells **flaccid** hoti hain, turgid nahi. Isliye yeh galat hai.

Q103. **(1)** Semi-lunar sclerenchyma patches wala pericycle **dicot stem** mein hota hai.

Q104. **(4)** Maize (monocot) ke bundles mein **water-containing cavities** (lysigenous) hoti hain.

Q105. **(1)** Dono sahi: umbilical cord placenta ko embryo se jodti hai. ICM mein pluripotent stem cells hoti hain.

Q106. **(2)** Guard cells ke paas wali specialised cells = **subsidiary cells**.

Q107. **(1)** Vascular bundles ring mein ⇒ **dicot stem**.

Q108. **(1)** Conjunctive tissue dicot root mein **xylem aur phloem ke beech** hota hai.

Q109. **(4)** Young dicot stem mein Trichome, Stomata, Pith hote hain. Bundle sheath nahi hota ⇒ **teen**.

Q110. **(4)** Meiosis-I ke baad primary spermatocyte se **secondary spermatocyte** banti hai.

Q111. **(1)** Mesophyll **parenchyma** se bana hota hai.

Q112. **(4)** Conjoint bundle mein xylem aur phloem ek hi radius par hote hain, aur ye stem/leaves mein common hain ⇒ **(1) aur (2) dono**.

Q113. **(4)** **Cortex** epidermal tissue system ka hissa nahi hai (ground tissue hai).

Q114. **(3)** Well-developed pith **monocot root** mein hota hai.

Q115. **(1)** Corners par cellulose, hemicellulose aur pectin se thick = **collenchyma**.

Q116. **(2)** Open bundle mein cambium hota hai, isliye secondary xylem/phloem ban sakte hain.

Q117. **(3)** Grasses mein intercalary meristem hota hai, isliye chare hue part regenerate ho jate hain. R, A ki sahi explanation hai.

Q118. **(3)** Monocot stem ki hypodermis **sclerenchymatous** hoti hai.

Q119. **(3)** Dicots mein guard cells **bean-shaped** hoti hain (monocots mein dumb-bell).

Q120. **(2)** Medullary rays = **radially placed parenchymatous cells**.

Q121. **(2)** Statement I galat (mesophyll vascular tissue nahi hai). Statement II sahi (spongy parenchyma palisade ke neeche hota hai).

Q122. **(2)** Starch sheath **endodermis** hai, epidermis nahi (A false). R true.

Q123. **(1)** Root/shoot tips ka meristem = **apical meristem**.

Q124. **(1)** Monocot stem ⇒ IV (peripheral bundles chhote). Root hairs ⇒ III (unicellular elongations). Mesophyll ⇒ II. Dicot stem ⇒ I (collenchymatous hypodermis).

Q125. **(1)** Casparian strips **endodermis** (cortex ki innermost layer) mein hoti hain aur waxy (suberin) hoti hain ⇒ only (a) and (b).

Q126. **(4)** Trichomes aam taur par multicellular, branched ya unbranched, aur secretory bhi ho sakte hain ⇒ **All of these**.

Q127. **(2)** I galat (monocot stem mein cambium nahi hota). II, III, IV sahi ⇒ **II, III, IV**.

Q128. **(1)** Cuticle **roots** mein absent hoti hai.

Q129. **(1)** Dorsiventral leaf ke 3 parts: **epidermis, mesophyll, vascular system**.

Q130. **(4)** Stomata ⇒ transpiration (III). Guard cell ⇒ stomatal pore ko enclose karta hai (I). Cuticle ⇒ waxy (IV). Trichome ⇒ multicellular (II).

Q131. **(2)** Sunflower (dicot) mein Casparian strips **root endodermis** mein hoti hain.

Q132. **(2)** Bundle sheath monocot stem mein hoti hai ⇒ **barley stem**.

Q133. **(3)** Lateral roots **pericycle** se shuru hoti hain.

Q134. **(1)** Protoxylem centre ki taraf ⇒ **endarch**.

Q135. **(1)** Dicot leaf mein **abaxial epidermis par stomata zyada** hote hain.

Q136. **(3)** Uterus ulte pear jaisa hota hai aur functional mammary gland sabhi female mammals ki characteristic hai ⇒ dono sahi.

Q137. **(4)** Baaki options mein sab single (1) ya sab paired (2) hain. (4) mein prostate/vagina single aur bulbourethral/ovaries paired hain ⇒ **odd one**.

Q138. **(2)** A sahi. B galat (15–20 lobes hote hain, 25–30 nahi). C galat (alveoli tubules mein khulte hain, aur tubules milkar duct banate hain, ulta likha hai).

Q139. **(3)** **LH** Leydig cells par act karke androgens ka synthesis aur secretion badhata hai.

Q140. **(2)** Vasa efferentia **rete testis se epididymis** tak sperm le jati hain.

Q141. **(2)** Meristem ⇒ actively dividing (III). Parenchyma ⇒ photosynthesis/storage (I). Epidermal ⇒ stomata (IV). Sclerenchyma ⇒ dead, no protoplast (II).

Q142. **(4)** Sperm-binding receptors **zona pellucida** par hote hain.

Q143. **(2)** **Prostate gland** unpaired hoti hai.

Q144. **(2)** Spermatogenesis ⇒ spermatogonia se sperm banna (II). Spermiation ⇒ sperms ka release (III). Spermiogenesis ⇒ spermatid se sperm (I). Sperm head ⇒ haploid nucleus (IV).

Q145. **(3)** Trichomes **transpiration se water loss rokte hain**.

Q146. **(4)** Seminal plasma mein fructose, calcium aur enzymes hote hain.

Q147. **(4)** Male external genitalia mein penis aur scrotum dono aate hain (I galat). Foetus 9 mahine mein poora develop hota hai (II galat).

Q148. **(2)** Graafian follicle phatne ke baad **corpus luteum** banta hai, jo **progesterone** secrete karta hai.

Q149. **(3)** First polar body **tertiary follicle** mein banti hai (meiosis-I ovulation se pehle poora hota hai).

Q150. **(3)** Medullary rays **ground tissue** ka part hain, vascular tissue ka nahi, isliye (3) galat hai.

Q151. **(3)** Acrosome sperm head ke **anterior** hisse par hota hai, isliye (3) galat hai.

Q152. **(4)** A, B, D sahi. C galat (urethra bladder se nikalti hai). E galat (myometrium smooth muscle hai).

Q153. **(4)** Menstrual cycle mein cyclic change **endometrium** mein hota hai.

Q154. **(3)** Fertilisation na hone par corpus luteum **degenerate** ho jata hai.

Q155. **(3)** Dono statements sahi.

Q156. **(2)** Fatty tissue ka cushion, skin aur pubic hair se dhaka = **mons pubis**.

Q157. **(4)** Placenta ⇒ oxygen/nutrient supply (IV). Implantation ⇒ endometrium mein embedding (III). Ovary ⇒ relaxin (II). ICM ⇒ embryo mein differentiate hone wale cells (I).

Q158. **(3)** **[?]** A sahi (ovary gamete banati hai). C sahi (dono taraf neeche pet mein). B galat maana gaya (ovary mainly steroid hormones banati hai). D galat (peripheral cortex aur inner

medulla hota hai) ⇒ **A and C**.

Q159. **(2)** Cyclic menstruation normal reproductive phase ka indicator hai.

Q160. **(4)** Monocot stem T.S. mein **D** (beech ka bada hissa) ground tissue hai. C vascular bundles hain.

Q161. **(3)** Sperm zona pellucida se contact karta hai (I sahi). Hymen virginity ka reliable indicator nahi hai (II sahi).

Q162. **(3)** **[?]** Female ke liye (a) aur (b) sahi hain. (c) galat (testes scrotum mein hote hain), (d) male cell hai ⇒ **Two**.

Q163. **(2)** Sperm head ke anterior end par cap jaisi structure = **acrosome**.

Q164. **(2)** Semen = **sperms** + **seminal plasma**.

Q165. **(4)** Follicular phase mein LH aur FSH dheere-dheere **badhte** hain, isliye (4) galat hai.

Q166. **(4)** Meiosis-II ke baad secondary spermatocyte se **spermatids** bante hain. Spermiogenesis mein spermatids se **spermatozoa** bante hain.

Q167. **(3)** Placenta hCG, estrogen, progesterone secrete karti hai (endocrine gland). R, A ki sahi explanation hai.

Q168. **(3)** Scrotum testes ko body temperature se 2–2.5°**C kam** rakhta hai.

Q169. **(2)** Vas deferens + seminal vesicle duct = **ejaculatory duct**.

Q170. **(4)** Males mein spermatogenesis puberty par shuru hota hai. Females mein oogenesis embryonic stage mein shuru ho jata hai.

Q171. **(4)** Hymen sudden fall, tampon, cycling jaise karanon se bhi toot sakti hai.

Q172. **(1)** Normal fertility ke liye kam se kam 60% sperms normal shape/size ke hone chahiye.

Q173. **(3)** A galat (zygote fallopian tube ke ampulla mein banta hai). B galat. C galat (meiosis-I ovulation se pehle poora hota hai). D aur E sahi.

Q174. **(3)** A, B, C sahi. D galat (endometrium menstrual phase mein toot-ta hai, follicular mein nahi).

Q175. **(3)** **Seminal vesicles** gland hain, baaki sab ducts hain.

Q176. **(2)** FSH aur LH **pituitary gland** banati hai.

Q177. **(1)** Secondary oocyte haploid hai, primary spermatocyte diploid hai.

Q178. **(4)** 100 secondary oocytes ⇒ 100 ova. 100 secondary spermatocytes ⇒ 200 sperms.

Q179. **(4)** Clitoris **labia minora** ke upper junction par hota hai.

Q180. **(3)** Spermatogonia mitosis se multiply hote hain aur phir spermatogenesis se sperms banate hain. Dono true aur R, A ki sahi explanation hai.